


# Backstage — Run and Deployment Guide

| Field | Details |
|---|---|
| Project | Backstage: Public Speaking Training App |
| Project No. | 8 |
| Teams | Frontend Group 7 and Backend Group 8 |
| Motto | Practice loud. Perform proud. |
| Guide scope | Local run, testing, and free Render/Vercel deployment |
| Source status | Existing application code; no functional change required |
| Speech engine | faster-whisper when available; mock fallback for lightweight demo hosting |

## 1. Requirements

- Python 3.10–3.13 recommended.
- Node.js 18 or newer for the frontend.
- Internet on the first backend setup for package/model downloads.
- Chrome or another modern browser for microphone recording.

## 2. Run the backend locally

Open a terminal in the project root:

```
cd backend
python run.py
```

The runner creates `venv`, installs dependencies, applies migrations, seeds prompts/settings, prepares Whisper when possible, starts Django, and opens the backend console.

Open:

- Console: `http://127.0.0.1:8000/`
- Admin: `http://127.0.0.1:8000/admin/`
- API: `http://127.0.0.1:8000/api/v1/`

If Whisper/model download is not available, use the safe lightweight mode:

```
python run.py --no-whisper
```

The UI will identify mock results. Mock mode does not transcribe the uploaded voice.

## 3. Backend checks

```
cd backend
python run.py test
python run.py check
```

Manual admin account:

```
python manage.py createsuperuser
```

Demo accounts:

```
python manage.py seed_demo --users
```

## 4. Run the frontend locally

Keep the backend running in one terminal. Open a second terminal:

```
cd frontend
npm install
npm run dev
```

Open `http://localhost:5173/`. The Vite development proxy forwards `/api` requests to the local Django server.

Frontend checks:

```
npm run build
npm run lint
```

## 5. Render backend deployment

Create a Render **Web Service** for the `backend` directory.

Build command:

```
pip install -r requirements.txt
```

Start command:

```
python manage.py migrate && python manage.py seed_demo --users && python manage.py runserver 0.0.0.0:$PORT
```

Set these private environment variables in Render:

```
DEBUG=0
SECRET_KEY=<long-random-secret>
ALLOWED_HOSTS=<render-backend-hostname>
CORS_ALLOWED_ORIGINS=https://<frontend-hostname>
TRANSCRIBER=mock
PROCESS_SYNC=0
WHISPER_DEVICE=cpu
WHISPER_COMPUTE_TYPE=int8
DEMO_ADMIN_PASSWORD=<new-private-admin-password>
DEMO_USER_PASSWORD=<new-private-user-password>
```

Use `TRANSCRIBER=mock` on a free Render instance because faster-whisper may exceed free CPU, memory, or model-storage limits. The full Whisper option is:

```
TRANSCRIBER=faster-whisper
```

Use it only on a host that can install `requirements-whisper.txt`, download the model, and run CPU transcription.

Check the deployed backend:

```
https://<render-backend-hostname>/
https://<render-backend-hostname>/admin/
https://<render-backend-hostname>/api/v1/
```

## 6. Free frontend deployment

For Vercel or Netlify, set the frontend project directory to `frontend`.

Build command:

```
npm run build
```

Output directory:

```
dist
```

For the full backend connection, set:

```
VITE_API_BASE=https://<render-backend-hostname>/api/v1
```

Then make sure the backend has this exact frontend origin:

```
CORS_ALLOWED_ORIGINS=https://<frontend-hostname>
```

## 7. Final run checklist

- Backend console opens.
- Frontend opens.
- User registration and login work.
- Prompt list loads.
- A 10-second-or-longer audio file can be selected.
- Upload returns processing status.
- Completed result shows pace, pauses, filler words, and transcript.
- Admin can open `/admin/`.
- `DEBUG=0` and a new `SECRET_KEY` are used on the deployed service.
- Real passwords and `.env` are kept private.

## 8. Deployment limitations

Free services may sleep when idle, reset local SQLite storage, and lack resources for real Whisper processing. The current application is suitable for academic/demo deployment. A production upgrade should use PostgreSQL, a durable queue, a production WSGI server, durable storage, HTTPS-only settings, and monitoring.